

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1204

Bairoa 3008 138 kV
November 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1204. The Interconnection Customer proposes a 51.2 MW solar photovoltaic facility interconnecting at the Bairoa 3008 138 kV substation in the Virginia area. The request was received on May 14, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 373 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$458,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Deltaport-Moonstone 138kV	Jim Hill Substation-Haggin 138kV	116.8	222	5.6
Jim ill Substation-Cedar Bayou Plant 138kV	Potosi-Tyler 138kV	105.6	236	5.8
Walshall-Yellow Creek Substation 138kV	N Tower Tap-Uknown 138kV	105.0	247	24.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Deltaport-Moonstone 138kV	Rebuild line with higher-capacity conductor	\$2,034,000	4% (\$88,000)
Jim ill Substation-Cedar Bayou Plant 138kV	Reconductor line	\$2,226,000	5% (\$111,000)
Walshall-Yellow Crk 138kV	Reconductor line	\$1,090,000	24% (\$259,000)
	Total allocated to project		\$458,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.